Solution Requirements

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/ Medium/Low)
1	Authentication	Disaster management officers and admin staff must securely log in using encrypted corporate credentials with mandatory Multi-Factor Authentication (MFA).	High
2	Authorization levels	The system must support role-based access: Admins control configurations, Disaster Officers issue alerts, and Field Responders can only view flood risk	High

		dashboards.	
3	External interfaces	The system must seamlessly connect to third-party Meteorological department APIs and regional IoT river-gauge sensor hardware data streams.	High

4	Transactions processing	The system must reliably handle, validate, and parse high-frequency real-time telemetry and data exchanges from thousands of live environmental sensors.	High
5	Reporting	The dashboard must display real-time predictive analytics graphs, flood risk heatmaps, and allow downloading automated historical data reports in PDF/CSV format.	Medium
6	Business rules, etc.	Proactive public evacuation alarms must only trigger automatically when live water metrics exceed critical predefined danger thresholds for 3 consecutive readings	High
7	Compliance to laws or regulations	System data handling and location logs must fully adhere to regional digital privacy standards and national disaster management security protocols.	Medium
8	Other	The system must maintain an unalterable operational audit log	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

	Performance & Speed	The system must parse raw input data streams and update predictive flood charts rapidly during live weather anomalies.	Dashboard charts must refresh within under 2 seconds of receiving raw telemetry.
	Scalability	The architecture must comfortably process heavy loads during sudden monsoon spikes without any system slowdowns.	Must easily support up to 50,000 concurrent sensor feeds and 1,000 active officer requests.
	Security & Data Privacy	All sensitive system operational credentials and emergency configuration settings must be fully secure from unauthorized access.	End-to-end data encryption using AES-256 standard for all network communications and logs.
	Reliability & Availability	The prediction engine and notification system must remain accessible and functional 24/7 throughout peak monsoon periods.	Guaranteed system uptime of 99.9% uptime per month

			during heavy rainfall months.
	Usability & Accessibility	The core warning interface must be clear, distraction-free, and accessible for dispatch officers under high stressful environments.	UI must comply with WCAG 2.1 AA standards for screen contrasts and simple navigatio n steps.
6. others		The system source code and database architecture must be clean, modular, and easy to deploy on local or backup cloud servers.	Maintainability:Zero downtime deplo Updates and under 30 minutes syste Disaster recovery time